

Lab Assignment-1

Communication Networks

L23+L24

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1) You are tasked with connecting two separate departments (HR and IT [two PCs for each department]) in a single building using Cisco Packet Tracer. Each department must be on its own subnet to reduce broadcast traffic. Outline the technical steps you would take to design, configure, and verify this setup.

Step 1: Design the Network

- Devices:
 - 1 Switch
 - 1 Router
 - 4 PCs (2 HR, 2 IT)
 - Subnets:
 - HR: 192.168.1.0/24
 - IT: 192.168.2.0/24
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Step 2: Build Topology in Packet Tracer

1. Add:
 - 1 Router (e.g., 2911)
 - 1 Switch (2960)
 - 4 PCs
 2. Connect:
 - PCs → Switch (straight-through)
 - Switch → Router
-

Step 3: Configure VLANs (on Switch)

```
enable
configure terminal
vlan 10
name HR
vlan 20
name IT
```

Step 4: Assign Ports to VLANs

Example:

```
interface fa0/1
switchport mode access
switchport access vlan 10
interface fa0/2
switchport mode access
switchport access vlan 10
interface fa0/3
switchport mode access
switchport access vlan 20
interface fa0/4
switchport mode access
switchport access vlan 20
```

Step 5: Configure Trunk Port (Switch → Router)

```
interface fa0/24
switchport mode trunk
```

Step 6: Configure Router (Inter-VLAN Routing)

```
enable
configure terminal
interface g0/0.10
encapsulation dot1Q 10
ip address 192.168.1.1 255.255.255.0
interface g0/0.20
encapsulation dot1Q 20
ip address 192.168.2.1 255.255.255.0
interface g0/0
no shutdown
```

Step 7: Assign IP Addresses to PCs

- HR PCs:
 - IP: 192.168.1.x
 - Gateway: 192.168.1.1
 - IT PCs:
 - IP: 192.168.2.x
 - Gateway: 192.168.2.1
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Step 8: Verification

- Ping within same department → should work
- Ping across departments → should work (via router)

2) Your

company occupies a three-story building with one 2960 Switch installed on each floor. There are two departments, Sales and Accounting, with employees distributed across all three floors. Using only Layer 2 Switches in Cisco Packet Tracer, how would you configure the network so that Sales members can communicate with other Sales members on different floors, while remaining completely isolated from the accounting department?

Step 1: Network Design

- 3 switches (one per floor)
 - VLANs:
 - VLAN 10 → Sales
 - VLAN 20 → Accounting
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Step 2: Create VLANs on ALL Switches

```
enable
configure terminal
vlan 10
name Sales
vlan 20
name Accounting
```

Step 3: Assign Ports

Example per switch:

```
interface fa0/1
switchport mode access
switchport access vlan 10
interface fa0/2
switchport mode access
switchport access vlan 20
```

Step 4: Connect Switches (Trunk Links)

Between floors:

```
interface fa0/24
switchport mode trunk
```

Do this on all inter-switch links.

Step 5: No Router Needed

- Since requirement = **isolation**
 - Do NOT configure inter-VLAN routing
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Step 6: IP Addressing

- Sales VLAN:
 - 192.168.10.0/24
- Accounting VLAN:
 - 192.168.20.0/24

Assign accordingly.

Step 7: Verification

- Sales → Sales (different floors) → works
- Accounting → Accounting → works
- Sales → Accounting → fails (correct behavior)